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Health Economics: Tools to Measure and Maximize Programme Impact

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1 Introduction

'Voluntary male circumcision can cut new HIV infections by half.' [1] This was the remarkable finding of a large clinical trial conducted in Orange Farm, South Africa, in 2005 [2]. Finally, a highly effective general population prevention strategy for HIV in Africa, with convincing biological outcome data! However, the health benefits of averted HIV disease are long-delayed while substantial costs are incurred now: voluntary medical male circumcision

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(VMMC) requires surgery, as well as risk reduction counselling and medical follow-up. Overall, is VMMC a good investment of limited public health funds? In 2006, we studied the cost of delivering VMMC, the gains in years of healthy life due to averted HIV infections, and the financial savings from unneeded lifetimes of HIV treatment. We found that VMMC costs much less than the averted health-care costs. Net savings; a very good investment indeed [3]. This information helped the World Health Organization (WHO) and other global agencies support widespread implementation of VMMC.

Global aspirations for health are ambitious, with the 2030 Sustainable Development Goals (SDGs) aiming to improve significantly on decades of progress in reducing disease burden [4]. Yet, after a period of rapid growth from 1997 to 2010, funding for global health is flattening largely due to a global economic slowdown. This divergence of mission and resources creates tensions for governments when choosing between evidence-based interventions to reach SDG targets. Economics can help resolve this contradiction. For example, SDG 3.4 states 'by 2030 reduce by one third premature mortality from non-communicable diseases through prevention and treatment'. Economics can inform an efficient combination of prevention (e.g. diet and exercise) and treatment (e.g. medications for hypertension and diabetes). Luckily, the tools of economics are increasingly sophisticated, and up to the task.

Economics uses and creates data that inform policy (Box 19.1). Economic analyses incorporate data from other public health disciplines, such as indices of disease prevalence and intervention efficacy. The analyses produce quantitative estimates of disease burden, costs, and cost-effectiveness which help policymakers prioritize diseases to target and choose between intervention strategies to maximize health gains with available resources.

Box 19.1 Policy-Relevant Questions That Economists Can Answer

- What does it cost to treat one person with a specified disease? Per cure? Per added year of life?
- What does it cost to deliver a prevention strategy? What is the estimated cost per case of disease averted?
- What is the most efficient balance of prevention and treatment strategies, for specific population groups and across the entire population?

We offer a broad review of health economics, focussing on methods to guide health-related resource allocation. We start with a brief discussion of economic systems, followed by a concise survey of major methods. We then explain two tools of economics most widely employed in global health: burden of disease metrics and cost-effectiveness analysis (CEA). We conclude with a brief discussion of the ethical basis for use of efficiency, as captured by CEA, to make decisions about prioritization in health.

2 Health Economics

Although some of its methods have been available for centuries, academics first characterized the discipline of *health economics* in the 1960s [5]. Since then, health economists have made significant contributions to public health decision-making. Just as our analyses cited earlier informed South African policymakers about the cost-effectiveness of introducing VMMC, others have, for example, demonstrated the cost-effectiveness of smoking cessation efforts [6] and assisted in the evaluation of national vaccination programmes [7], long-lasting insecticide impregnated bednets and other strategies to prevent malaria [8].

Health economists view health systems as economic systems in which health-care and public health providers are the suppliers and individuals the consumers (and presumed beneficiaries), each making choices with limited resources and with goals of maximizing income or health status for themselves or for populations. Policymakers intervene in the system to achieve societal and political goals. Economists have the tools to assess these dynamics and measure the resulting effects on populations and individuals, for small and larger portions of the system. When decision-makers consider alternative public health policies, they are often interested in how the respective implementation costs of these policies compare with expected benefits to society. Whereas other researchers propose and demonstrate the health benefits of specific interventions, economists consider the costs of actually implementing these interventions and the expected benefits in real-world operation.

Health economists draw on the traditional arsenal of economic methods such as econometrics, micro-costing and cost-benefit analysis (CBA) and have developed approaches specific to the health sector. The range of methods which health economists combine or apply include:

Econometrics This statistical arm of economics aims to reveal and quantify causal relationships that drive economic systems, such as socio-demographic factors that determine utilization of medical services, lifestyles that influence longevity, or programme design features that determine intervention cost. Several econometric methods (e.g. instrumental variables and regression discontinuity design) are especially effective at confirming causality by ruling out self-selection and other reverse causality (known as endogeneity). An excellent example of econometrics is an analysis of the health effects of foreign assistance in health. Bendavid et al. applied econometric methods to cross-national panel data on health aid and health – as summarized by life expectancy at birth and under-5 mortality – in 140 LMICs. The study found that 'Foreign aid to the health sector is related to increasing life expectancy and declining under-5 mortality. The returns to aid appear to last for several years and have been greatest between 2000 and 2010, possibly because of improving health technologies or effective targeting of aid.' [9]

Cost-Benefit Analysis CBA, which derives from the academic field of social welfare, assesses if a particular economic activity is worth while in monetary terms. CBA can be used to demonstrate the economic returns of investment in an intervention, to compare the costs and benefits of alternative interventions, and to help policymakers allocate budgets. CBA compares the costs of production (e.g. of health services) with the monetary value that society places on the outcomes (e.g. the value of averted costs in health and other areas, or willingness to pay for improved health status). Keen et al., for example, compared the costs of family planning (FP) delivery in Sierra Leone with savings from five social services (primary education, child immunization, malaria prevention, maternal health services, and improved drinking water) [10]. They projected that, with high access to FP, the population would reach 8.3 million by 2035 versus 9.6 million with no FP expansion. They estimated a US$2.10 saving in social services for each dollar spent on FP, representing a 2:1 benefit-cost ratio. But Keen et al. also pointed out that there are other important health benefits associated with scaling up FP such as reduced maternal and child mortality that are not included in the benefit-cost calculation and that some benefits, such as improved women's rights and gender equity 'are more appropriately assessed qualitatively'.

Cost-Effectiveness Analysis CEA, used widely in health, assesses the cost of an intervention per unit of health gained. The *cost* component (the numerator) is the same as cost in CBA. However, the *effectiveness* component (the denominator) is expressed in health units: clinical events (e.g. deaths or illness episodes), other health outcomes (e.g. new infections), or, ultimately and preferred, a metric that quantitatively combines morbidity and mortality, such as Disability-Adjusted Life Years (DALYs). We explain DALYs in Sect. 3 and CEA in depth in Sect. 4.

Micro-costing Micro-costing quantifies resources and associated costs needed to deliver a set of services, such as a prevention or treatment intervention. Micro-costing is the standard of practice to characterize costs usually from the bottom up, and permits an excellent view into the production process. An important distinction: *price* is what buyers pay to sellers for a service or commodity, whereas *cost* represents the amount put into production. The difference is profit. An excellent example of micro-costing is from the ORPHEA study of HIV interventions in hundreds of facilities in multiple countries [11].

Behavioural Economics This discipline merges cognitive psychology and economics. It uses insights from the study of human motivation to design strategies to foster healthier behaviours or better clinical practices. Specific topics include the role of mental short cuts (our quick instinctive reactions to situations we encounter), framing (how a choice is posed), and incentives (e.g. monetary rewards or internal norms). A 2017 review of behavioural economics to encourage physical activity in patients found that tools such as precommitment contracts and framing are valuable in this setting [12]. A Kenyan field trial of bednets to prevent malaria examined if paying for (in behavioural economic terms, *investing in*) nets affects use [13]. It found that cost-sharing (as opposed to free nets) does not improve use but does reduce demand, suggesting the greater health impact of free distribution. A later Cochrane systematic review confirmed the consistency of these findings across studies [14].

Discrete Choice Experimentation DCE elicits preferences about potential actions (e.g. choice of a health clinic). Methods to explicitly ask individuals about their preferences may be inaccurate, distorted by social desirability (saying what is perceived as acceptable) and other cognitive biases. Instead, the discrete choice approach *observes* the choices, and then uses statistical techniques to infer what attributes (e.g. cleanliness, politeness, or drug stock) drive the choices. Thus discrete choice methods reveal actual rather than stated reasons for choices. Use of the approach in Ethiopia and Mozambique found that to retain women in lifelong HIV care, the important attributes were respectful provider attitudes and ability to obtain non-HIV health services during HIV-related visits. Facility type, that is hospital versus health centre, was less important [15].

Financing Financing addresses how funds are raised, for example, from taxes and insurance premiums, and how they are distributed to providers, for example, via fee-for-service or capitation. Thus, financing characterizes how money moves through, and lubricates, the economic system. Comprehensive ongoing work led by Joseph Dieleman at the Institute of Health Metrics and Evaluation (IHME) documents domestic and international health financial flows by source and mechanism, country, and disease area [16].

Labour or Workforce Economics This area of inquiry describes the professionals, for example, doctors, clinical officers, and nurses, who perform the services required for the health system to function. This is the biggest (most costly) supply component of health systems. Projections of workforce need and training are critical to prepare for future health-care needs. An analysis of physician production and supply in Tanzania examined the potential impact of interventions to retain physicians in training and practice [17]. The World Health Organization (WHO) just launched a new initiative on health workforce development to pursue the Sustainable Development Goals (Chap. 12) [18].

Health economists use a combination of tools to apply economics to pressing challenges in global health. For example, an effort to define the optimal mix of interventions might rely on DALYs to quantify disease burden at baseline and after intervention, micro-costing to assess the resources required for alternative strategies, and CEA to assess health value for money. A study of a local market for primary medical care could use discrete choice experiments to understand patient preferences among providers with different traits, and econometrics to assess how those traits relate to actual utilization.

3 Measuring Burden of Disease

For economic analysis, a preferred measure of health benefits is one that can be compared across locations, over time, and by disease categories. Some studies use summary measures of health status such as life expectancy, infant and child mortality; others use disease-specific disease incidence or prevalence. Ideally what is needed is a single summary metric that comprises health gains encompassing both deaths averted and reductions in adverse non-fatal health states. Work in this area started during the 1970s with the QALY (Quality-Adjusted Life Year) and the DALY which was introduced in the 1990s. The DALY and the QALY are the converse of each other: DALYs measure burden of disease and QALYs measure health. The DALY is the metric used for the development of global and national/sub-national burden of disease estimates. IHME publishes the Global Burden of Disease (GBD) Study, and the WHO produces similar estimates [16, 19].